

# STPWR: A Spatiotemporal Prediction-based Worker Pre-Recruitment Framework for Mobile Crowd Sensing

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## Abstract

Mobile Crowd Sensing (MCS) harnesses the power of participatory data collection to address pressing societal challenges, from urban environmental monitoring to disaster response coordination. Conventional methods typically recruit workers only after a task is generated, leading to significant response delays, especially for time-sensitive tasks. To address this limitation, we propose the Spatiotemporal Task Prediction-based Worker Pre-Recruitment (STPWR) framework. First, we tackle data sparsity by developing a trajectory-filling technique that leverages movement and behavioral similarities among task publishers. Using these enriched data, we design an LSTM-based prediction model to predict future task distributions with high accuracy. Second, we formulate worker pre-recruitment as a multi-objective optimization problem, balancing task completion rate, worker travel distance, and response time. A genetic algorithm is employed to derive an efficient recruitment strategy. We evaluate STPWR on a real-world dataset (Feeder Taxi GPS Data), comparing it against baseline methods including Gibbs-Inspired LSTM, Priority Greedy, and OTAGA. Experimental results demonstrate that STPWR improves task prediction accuracy by 15–20%, achieves a task completion rate of 80–95%, and reduces

response time by 30% compared to existing approaches. More importantly, STPWR reduces total worker movement distance by up to 25%, translating to substantial energy savings and underscoring its potential as a sustainable MCS solution. These findings validate the effectiveness of our pre-recruitment strategy in building efficient and environmentally conscious MCS systems for social good.

## CCS Concepts

• **Computer systems organization** → **Sensor networks**; • **Human-centered computing** → **Ubiquitous and mobile devices**; **Collaborative and social computing**.

## Keywords

Mobile Crowd Sensing, Worker Pre-recruitment, Spatiotemporal Prediction, Genetic Algorithm

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## 1 Introduction

With the popularity of intelligent mobile devices equipped with diverse sensors, mobile crowd sensing (MCS) has become an effective sensing paradigm [1, 9, 20, 22, 28] and has been widely applied in various areas, such as collecting real-time air quality data for pollution alerts, monitoring dynamic traffic flow for congestion

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management, and detecting water pollution sources promptly [8, 11, 15, 19, 32]. In MCS, the worker recruitment method has been researched as a very important issue, as it directly influences the efficiency of task execution [4, 7, 10, 17, 18]. Beyond technical efficiency, MCS has a profound potential to contribute to the United Nations Sustainable Development Goals (SDGs), such as creating sustainable cities and communities and taking urgent action to combat climate change. For instance, MCS can be deployed for dynamic traffic flow management to reduce congestion emissions, for real-time air quality mapping to protect public health, and for rapid damage assessment to coordinate disaster relief efforts.

The existing worker recruitment methods are mainly divided into two categories: the first is to recruit according to the location of the task, such as when the task comes, considering the time and place of the task, workers are recruited to improve the task completion rate [5]. Second is to recruit according to the trajectories of workers, for example, by sensing real-time location and utilizing historical trajectory data of workers, it becomes feasible to predict the future activity area of these workers accurately, then workers can be strategically recruited for tasks [3, 16]. However, the above methods mainly serve tasks that have already occurred, and cannot perform worker pre-recruitment before task occurrence. Thus, they neglect the effect of task distribution predictability, and have not revealed that, when the task has strict timeliness, worker pre-recruitment is a feasible method to reduce task response time.

Based on the above findings, this paper proposes a Spatiotemporal Task Prediction-based Worker Pre-recruitment (STPWR) method. Unlike conventional approaches that recruit workers only after tasks appear, STPWR proactively recruits workers by predicting spatiotemporal task distributions, thereby significantly reducing task response time. As illustrated in Figure 1, publishers  $P_1$  and  $P_2$  are predicted to generate tasks in cells  $C_2$  and  $C_3$ , respectively, enabling the platform to pre-assign workers accordingly. This pre-recruitment paradigm, however, introduces two key challenges.

The first one is the sparsity of historical data. When predicting task distribution, the incompleteness of historical data of task publishers greatly reduces the accuracy of task distribution prediction. This makes the recruited workers deviate from the actual needs, and then affects the task completion rate.

The second one is the trade-off between moving distance and response time. When recruiting workers, in order to reduce the response time, there is a tendency to assign workers who are closer in distance to perform tasks quickly. However, this is often accompanied by frequent movement of workers, resulting in an increase in the moving distance. On the issue of data sparsity in trajectory prediction, in [25], a time-variant Markov model based on Gibbs sampling has been designed for mobility prediction, to solve the problem of high missing rate, in which the Gibbs sampling method is used to recover the missing parts of a trajectory.

Advanced predictive models now integrate structural priors to enhance accuracy. For instance, Kang et al. [13] introduce StrucFormer, a structural prior-guided Transformer designed for high-fidelity data inference in sparse mobile crowdsensing scenarios. In [26], a multi-task learning algorithm has been designed to predict the sparse trajectories and improve the prediction performance, which learns the similarity of different users. However, these methods primarily focus on the trajectory itself and fail to fully predict

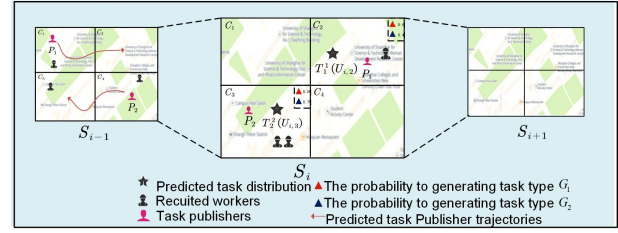


Figure 1: A scenario of work recruitment.

the spatiotemporal distribution of tasks closely related to the trajectories.

On the issue of worker travel distance and task response time, in [12], considering that the task is related to a specific location, workers need to go to the designated cell and perform the task. By optimizing the moving distance of workers, the travel distance is effectively reduced. In [2], by planning the movement path of workers and taking into account the task deadline at the same time, the movement distance of workers is effectively reduced. In [34], the recruitment of workers is carried out by obtaining the future movement trajectory of workers, aiming at effectively shortening the task response time. However, these methods do not consider the future distribution of tasks and the current recruitment status of workers, making it impossible to effectively strike a balance between task response time and mobility efficiency.

This paper introduces a method for worker pre-recruitment based on spatiotemporal task prediction. The method addresses the challenge of sparse historical data and predicts the spatiotemporal distribution of tasks, thereby reducing unnecessary movement of workers and ensuring that tasks receive prompt responses. The key contributions of this paper are outlined as follows:

- A Spatiotemporal Prediction Model with Trajectory Filling: We address data sparsity by leveraging similarities in movement and behavior among task publishers to fill missing historical data, and employ an LSTM-based model to accurately forecast future task distributions.
- A Multi-Objective Pre-Recruitment Optimization Framework: We formulate worker pre-recruitment as a multi-objective optimization problem, balancing task completion rate, worker travel distance, and response time, and solve it efficiently using a genetic algorithm.
- Extensive Validation and Performance Gains: Comprehensive experiments on a real-world dataset demonstrate that our framework improves prediction accuracy by 15-20%, achieves high task completion rates of 80-95%, and reduces both response time and worker movement by up to 25%.

## 2 Related Work

### 2.1 Worker recruitment methods in MCS

The existing methods for recruiting workers can be mainly divided into two categories: recruitment based on the location of tasks and recruitment based on the trajectories of workers. For recruitment based on the location of tasks, for example, in [6], a service framework was designed based on location-aware crowdsensing systems with time constraints, aiming to efficiently recruit workers for GPS

location-based tasks to ensure high-quality results. In [21], they combined the location and time window of the task to design a profit-oriented worker recruitment method that ensures the worker completes the task before the deadline. In terms of collaborative task management, [35] proposes a privacy-preserving hybrid multi-task allocation scheme (PPHMA) that optimizes the coordination among diverse workers while ensuring the security of sensing data in complex multi-task environments.

For recruitment based on the trajectories and preferences of workers, in [3, 14, 16], they predicted worker mobility and performed recruitment after task generation, effectively matching workers to the designated tasks while minimizing deviations from their daily trajectories. This approach significantly optimizes the overall task completion rate. In [3], the grid-based prediction method was designed to estimate the future location distribution of workers by prediction, taking into account the spatial characteristics of current workers and tasks, and achieving the maximization of the global worker recruitment quality under the travel cost constraint. In [27], a geography-based model predicted the probability of workers passing through specific areas at certain times, focusing on minimizing the incentive budget for task allocation. Additionally, in [16], future movement trajectories of workers were considered, along with changes in task distribution across different locations and sensing cycles, ensuring that tasks were allocated without deviating from the workers' established routes. In [17], a stable task assignment mechanism is introduced for multi-platform MCS environments, emphasizing the importance of matching robustness and long-term stability in worker recruitment under cost constraints.

However, in the above methods of worker recruitment, recruitment often begins after the task is generated. Recruiting workers only after the task is generated will increase task response delay. Therefore, it is crucial to develop an effective pre-recruitment method for workers to reduce the response delay of tasks.

## 2.2 Trajectory prediction in MCS

When task publishers generate tasks and need to recruit workers, the task generation is closely linked to the trajectory of the task publishers. Therefore, the role of trajectory prediction in worker recruitment is crucial. In [32], a subarea evaluation method based on deep reinforcement learning is developed for sparse mobile crowd sensing (SMCS), specifically addressing worker mobility and sensing cycle constraints to minimize data inference errors in urban environmental sensing. In [24], a mobile crowd-testing worker recruitment strategy is presented, employing a POI-based mobility prediction approach similar to the semi-Markov model, primarily focusing on optimizing data upload costs. In [23], the semi-Markov model is employed to predict the probability of workers appearing at specific locations by translating their movements into transitions between cells. In [30], the context-aware trajectory prediction method utilizing two types of LSTM networks is designed, transforming real-valued target trajectories into discrete path segments within a gridded system.

In trajectory prediction, data sparsity primarily presents itself as a discontinuity or absence of trajectory data over time and space. This lack of data poses a challenge for prediction methods, which depend on having complete historical information.

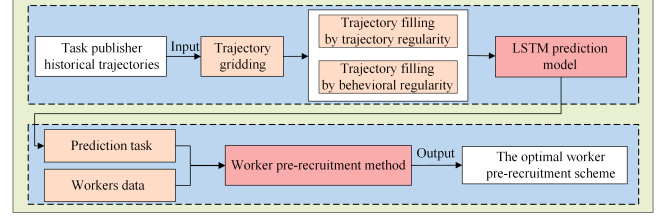


Figure 2: System Model

When using trajectory prediction methods, the main purpose is to obtain the spatiotemporal distribution of tasks generated by the publisher, and conduct pre-recruitment of workers, thereby shortening response time. In considering the publisher's trajectory information, it is necessary to consider the similarity of movement trajectories and the behavioral patterns they exhibit in publishing tasks. Patterns include task type, generation frequency, time, and location, which are essential for understanding their requirements and predicting behavior. Although motion trajectories provide geolocation information, behavioral patterns are not adequately considered. Therefore, to more accurately predict and respond to the publisher's requirements, a comprehensive analysis of behavioral patterns and motion trajectories is required.

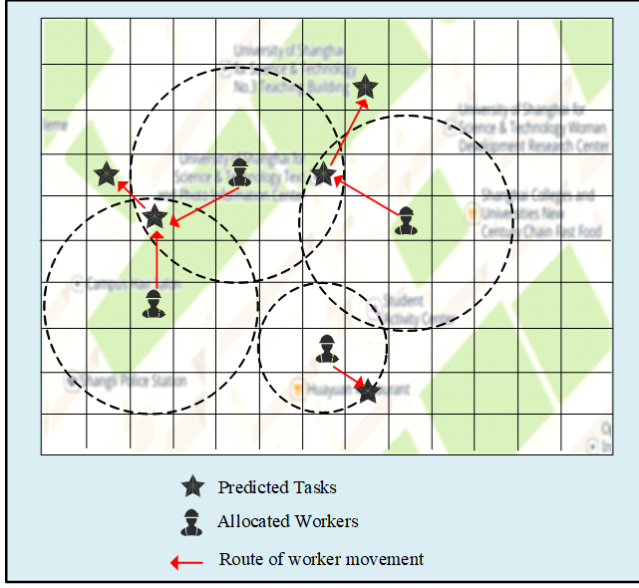
## 3 System Work

To formalize the interactions and computations within the proposed system model, we summarize the key mathematical notations in Table 1 in the Appendix. These symbols are foundational to the problem formulation and the design of our STPWR method. The sets  $A, C, S, W$  and  $G$  define the core entities of the MCS environment. Critical properties such as worker execution capacity  $Cap(W_k)$ , task requirement  $Cap(T_n^h(U_{i,m}))$ , and worker preference  $Pref(W_k)$  directly influence the recruitment strategy. The probabilistic measures  $P(A_h)(U_{i,m})$  and  $P_n^h(G_n)(U_{i,m})$ , predicted by our LSTM model, quantify the future likelihood of task generation. Furthermore, the similarity metrics  $Jaccard_{h,l}$  and  $LCSS_{tr_h, tr_l}$  are central to our data filling technique. The decision variable  $X_k(T_n^h(U_{i,m}))$  and the objective measures  $Tcr, WaitTime$  and  $Dis(W_k)$  are instrumental in formulating the optimization problem presented in the following section.

### 3.1 System Model

As illustrated in Figure 2, the system model includes task publishers, a platform, and workers. The platform gathers historical data from task publishers and uses this information to predict task generation. For each predicted task, the platform is responsible for pre-recruiting workers. Once the workers are pre-recruited by the platform, they will proceed to the designated location where the task is generated to carry out the task.

**Task environment:** The task environment is partitioned into cells. The set of cells is denoted by  $C = C_1, C_2, \dots, C_m, \dots, C_M$ . Each day is divided into time slots, and the set of time slot is denoted by  $S = S_1, S_2, \dots, S_i, \dots, S_I$ . A pair of cell and time slot is called a spatiotemporal unit. For instance,  $U_{i,m}$  represents the spatiotemporal unit within the cell  $C_m$  at time slot  $S_i$ . and  $(x(U_{i,m}), y(U_{i,m}))$  is the coordinate of the center point of the  $U_{i,m}$ .



**Figure 3: Worker movement trajectories during worker recruitment.**

**Task publishers:** In the above environment, there is a set of task publishers  $A = A_1, A_2, \dots, A_h, \dots, A_H$ . Each task publisher may generate multiple types of tasks, denoted by  $G = G_1, G_2, \dots, G_n, \dots, G_N$ . And the task generated by task publisher  $A_h$  in unit  $U_{i,m}$  with task type  $G_n$  is denoted by  $T_n^h(U_{i,m})$ , the execution capacity required by  $T_n^h(U_{i,m})$  is denoted by  $Cap(T_n^h(U_{i,m}))$ .

**Workers:** There are a set of workers  $W = W_1, W_2, \dots, W_k, \dots, W_K$  responding for recruitment from the platform. For worker  $W_k$ , its execution capacity, task type preference and current spatiotemporal unit are denoted by  $Cap(W_k)$ ,  $Pref(W_k)$  and  $Unit(W_k)$ , and its task response range is  $SCO(W_k)$ , and it only accepts the recruitment of tasks in its task response range.

**Platform:** The platform is responsible for collecting information of task publishers, predict the spatiotemporal distribution of tasks that will be generated by these task publishers, then pre-recruit workers for these tasks.

### 3.2 Problem Formulation

In our worker pre-recruitment framework, the primary goal is to devise a recruitment strategy that optimizes several competing objectives. We define these objectives as follows:

**Task completion rate (TCR):** The TCR measures the ratio of the total execution capacity assigned to a task versus the capacity it requires. It is defined as:

$$Tcr(T_n^h(U_{i,m})) = \frac{Cap'(T_n^h(U_{i,m}))}{Cap(T_n^h(U_{i,m}))}, \quad (1)$$

where  $Cap'(T_n^h(U_{i,m}))$  denotes the sum of the execution capacity of all assigned workers, calculated as:

$$Cap'(T_n^h(U_{i,m})) = \sum_{k=1}^K X_k(T_n^h(U_{i,m})) \cdot Cap(W_k) \cdot Pref(W_k), \quad (2)$$

where  $X_k(T_n^h(U_{i,m}))$  is the flag denoting whether worker  $W_k$  is assigned to task  $T_n^h(U_{i,m})$ .

**Task response time:** The task response time is the time interval from task generation time to the start time of task execution.

**Worker movement distance:** The worker movement distance is the distance between the current location and task execution location. We use Manhattan distance to simulate real environment.

$$Dis(W_k) = |x(U_{i+1,m}) - x(U_{i,m})| + |y(U_{i+1,m}) - y(U_{i,m})|. \quad (3)$$

**Worker task response range:** The task response range of workers is the area where workers can respond. Beyond this range, workers cannot be recruited.

**Task priority:** Task priority is a reflection of task response time. The longer a task remains, the higher its priority.

$$Pri(T_n^h(U_{i,m})) = 1 - \frac{1}{WaitTime(T_n^h(U_{i,m}))}, \quad (4)$$

if  $WaitTime(T_n^h(U_{i,m})) > 0$ .

To ensure high-quality task completion while effectively minimizing the distance that workers need to travel and reducing response times, the goal is to optimize both the task completion rate and the distance traveled by workers. This dual optimization will be achieved through a specific modeling approach.

$$\begin{aligned} & \text{maximize} \left( \sum_{j=1}^J Tcr(T_n^h(U_{i,m})) - Pri(T_n^h(U_{i,m})) - \right. \\ & \quad \left. \sum_{i=1}^I \frac{Dis(W_k)}{row^{max} + col^{max}} \right), \quad (5) \\ & 0 \leq Tcr(T_n^h(U_{i,m})) \leq 1, \quad \forall h, n, i, m, \\ & 0 \leq Dis(W_k) \leq SCO(W_k), \quad \forall k. \end{aligned}$$

The specific implementation algorithm for the workers' pre-recruitment model is described in Section IV of our research.

## 4 Method

To clearly illustrate the spatiotemporal task prediction-based worker pre-recruitment method, we begin by devising an LSTM-based task distribution prediction model. Following this, we utilize a genetic algorithm to design an optimal worker pre-recruitment strategy.

### 4.1 Task prediction

To predict the task distribution, there are three sub-steps, including trajectory gridding, trajectory filling and task prediction based on LSTM.

**Trajectory gridding:** Trajectory gridding is to convert a task publisher's historical GPS sequence to a cell sequence. For example, a task publisher  $A_h$  historical GPS sequence is denoted as  $LS_u = (S_1, L_1), (S_2, L_2), \dots, (S_i, L_i)$ , where  $L_i$  is the coordinate (longitude, latitude) indicating the task publisher  $A_h$  historical location at time slot  $S_i$ . For instance, in Figure 4,  $A_h$  historical GPS sequence  $(S_1, L_1), (S_2, L_2), \dots, (S_i, L_i)$  is converted to the spatiotemporal cell sequence (each element is represented as  $(S_1, C_1), (S_2, C_2), \dots, (S_i, C_m)$ ) via trajectory gridding.

Table 2 shows the trajectory results of several task publishers after trajectory gridding, for example, task publisher  $P_1$  located in

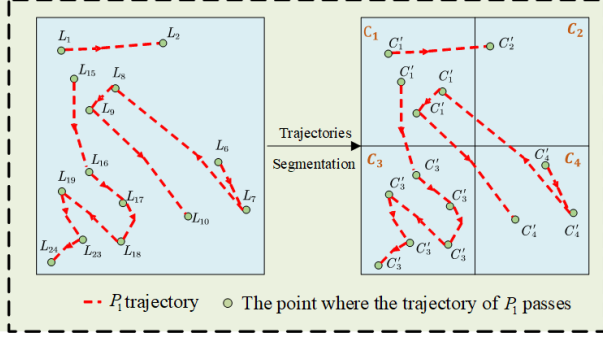


Figure 4: Schematic diagram of trajectory gridding.

cell  $C_1$  at time slot  $S_1$  in day  $D_1$ . The missing elements are due to the lack of task publisher locations information at the corresponding spatiotemporal units. Table 2 is detailed in the Appendix.

Figure 3 shows the trajectory gridding of task publisher  $P_1$ . The historical data has been organized into a grid corresponding to the locations where task publishers are situated at each moment.

**Trajectory filling:** To address the challenge of sparse historical data, we employ a comprehensive trajectory-filling strategy that leverages two complementary sources of information: the intrinsic regularity of individual movement patterns and the behavioral commonalities across different task publishers. Firstly, by analyzing the daily trajectory regularity of a single task publisher, missing location information in a specific spatiotemporal unit can be inferred from known data in corresponding units of previous days or from adjacent units within the same day, ensuring the filled trajectory maintains spatiotemporal smoothness and consistency with the publisher's overall movement profile. Secondly, we recognize that task publishers with similar behaviors and trajectories are likely to visit comparable locations. Therefore, we assess inter-publisher similarity by examining both their historical task publication records and their movement trajectories. For a publisher with missing data, we can accurately impute the gaps using the known trajectory information from the most behaviorally analogous publishers. This dual-strategy approach significantly enriches the dataset, providing a more robust foundation for subsequent task prediction and directly enhancing the model's capability to serve areas that might otherwise be underrepresented due to data scarcity.

The behavior similarity between task publisher  $A_h$  and publisher  $A_l$  is represented by  $Jaccard_{u,v}$  and it can be calculated as:

$$Jaccard_{h,l} = \frac{|I_h \cap I_l|}{|I_h \cup I_l|}, \quad (6)$$

where  $I_h$  is the set of tasks published by publisher  $A_h$ , and  $I_l$  is the set of tasks published by publisher  $A_l$ .

The trajectory similarity between trajectories of task publishers  $A_h$  and  $A_l$  is denoted by  $LCSS_{tr_h, tr_l}$  and it can be calculated as:

$$LCSS_{tr_h, tr_l} = \begin{cases} 0 & , \text{ if } tr_h = \emptyset \text{ or } tr_l = \emptyset, \\ 1 + LCSS_{rest(tr_h), rest(tr_l)} & , \text{ if } d(head(tr_h), head(tr_l)) \leq \epsilon, \\ \max(LCSS_{rest(tr_h), tr_l}, LCSS_{tr_h, rest(tr_l)}) & , \text{ otherwise.} \end{cases} \quad (7)$$

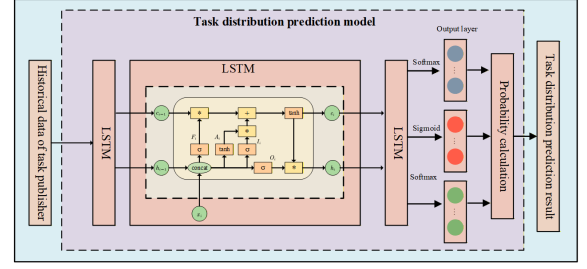


Figure 5: The LSTM task prediction model structure.

where  $tr_h$  is the task publisher  $A_h$ 's trajectory and  $tr_l$  is task publisher  $A_l$ 's trajectory.  $head(tr_h)$  is the first element in task publisher  $A_h$ 's trajectory, and  $rest(tr_l)$  is the remaining trajectory of  $A_l$  without the first element. The parameter  $\epsilon$  is a defined threshold.

The total similarity between task publisher  $A_h$  and task publisher  $A_l$  is

$$Similar_{h,l} = \eta \cdot Jaccard_{h,l} + (1 - \eta) \cdot LCSS_{tr_h, tr_l}, \quad (8)$$

where  $\eta \in [0, 1]$  is a hyper parameter.

By leveraging similarities among task publishers, missing location data for a task publisher in a spatiotemporal unit of a day can be filled by known location data of the most analogous publisher.

**Task prediction based on LSTM:** To learn features from historical data of task publishers, the LSTM is used to predict future task generation.

The output of the prediction model is divided into three layers to represent the probability sequences related to task generation. As shown in Figure 5, the initial layer operates a softmax function to output probabilities of a task publisher to visit future spatiotemporal units. The visitation probability of task publisher  $P_h$  to spatiotemporal unit  $U_{i,m}$  is represented as  $P(A_h)(U_{i,m})$ . The second layer operates as a binary classifier, indicating to the task publisher whether to generate a task in the predicted spatiotemporal unit. The third layer, conditioned on the prediction from the second layer, uses a softmax function to output the probability of different task types, which are only valid if the second layer confirms task generation. The probability of task publisher  $P_h$  generates a task with type  $tp_n$  based on predicted spatiotemporal unit probability  $P(A_h)(U_{i,m})$  is represented by  $P(P_n^h(G_n)(U_{i,m})|P(A_h)(U_{i,m}))$ , which is described in detail as follows.

$$\begin{aligned} & P(P_n^h(G_n)(U_{i,m})|P(A_h)(U_{i,m})) \\ &= P(P_n^h(G_n)(U_{i,m})|P(T^h(U_{i,m}))) \cdot P(P(T^h(U_{i,m}))|P(A_h)(U_{i,m})), \end{aligned} \quad (9)$$

where  $P(T^h(U_{i,m}))$  is the probability of publisher  $A_h$  generating a task.

## 4.2 The spatiotemporal task prediction-based worker pre-recruitment method

In this section, we present a genetic algorithm developed to address the issue of pre-recruiting workers based on predictions of task distribution. In this algorithm, workers are treated as genes within chromosomes. The goal is to create a chromosome that maximizes the task completion rate while minimizing task response time

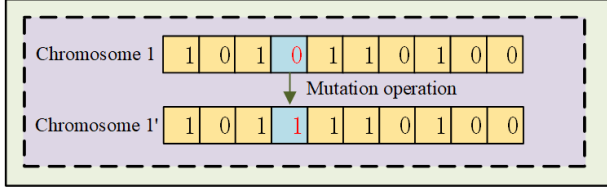


Figure 6: Mutation operation.

and the distance workers need to travel. The algorithm consists of several components: population initialization, chromosome representation, fitness function, selection operation, crossover operation, and mutation operation, each of which is detailed below.

**Population initialization:** The worker pre-recruitment method presented assigns a set of workers to each predicted task. To achieve a uniform distribution of feasible solutions throughout the solution space, we initialize the population randomly. For a task, the designed algorithm selects a number of workers from the candidate workers set  $W$ , and repeats this process until an initial population comprising  $N$  chromosomes are formed.

**Chromosome representation:** In this paper, a chromosome represents a worker pre-recruitment scheme. Table 3 provides a detailed exposition of the chromosome binary encoding scheme. The scheme assigns task  $T_n^1(U_{i,m})$  to the workers  $W_3, W_4, W_8, W_9$ . Table 3 is detailed in the Appendix.

**Fitness function:** In this paper, the objective is to maximize the task completion rate, based on which, another optimization objective is to minimize the movement distance of workers. Since fitness is directly proportional to the task completion rate and inversely proportional to movement distance, the fitness function can be expressed as:

$$Fitness = Tcr \left( T_n^h(U_{i,m}) \right) - Pri \left( T_n^h(U_{i,m}) \right) - \frac{Dis(W_k)}{row^{max} + col^{max}}. \quad (10)$$

**Selection operation:** In this paper, a roulette wheel selection method is designed to execute the following selection operations: 1) Calculate the total fitness. 2) Calculate the selection probability of each chromosome based on its proportion of the total fitness. 3) Select chromosomes according to their probabilities. For a population with  $R$  chromosomes, the selection probability for the  $i$ -th chromosome can be calculated as:

$$Pro(i) = \frac{Fitness_i - Fitness_{min}}{\sum_{j=1}^R (Fitness_j - Fitness_{min})}. \quad (11)$$

**Crossover operation:** To achieve the optimization objectives while ensuring that each worker is assigned to only one area at a time, the order crossover (OX) operation is selected. We employ a crossover probability of  $P_c = 0.85$ .

**Mutation operation:** The mutation operation is achieved by adding or removing worker genes from the chromosome. The mutation probability is set to  $P_m = 0.01$ . For example, as shown in Figure 6, a new gene  $w_4$  is added to chromosome 1. It is worth noting that since a worker is allocated to a task at most once, it is necessary to avoid adding a worker who already exists in the chromosome. In addition, for a chromosome that contains only one worker, removing the worker is also not allowed. Based on the

above concepts, the steps of the STPWR can be described as follows: i) Generate an initial population with  $R$  chromosomes, where each chromosome represents a worker recruitment scheme. ii) Calculate the fitness of all chromosomes in the population using the designed fitness function. iii) Generate a new generation population through selection, crossover, and mutation operations. iv) Repeat steps ii) and iii) until the maximum number  $Y$  of iterations is reached, and output the chromosome with the highest fitness as the optimal solution. These steps are detailed in Algorithm 1 in the Appendix.

## 5 Experimental Evaluations

### 5.1 Experimental Setting

**Dataset and Model Settings:** In this paper, the data set Feeder [33] is used, which encompasses five distinct types of data, Cellphone CDR Data, Smartcard Data, Taxicab GPS Data, Bus GPS Data, and Truck GPS Data, all sourced from Shenzhen, a city in China. In our study, we focus specifically on the Taxicab GPS Data due to its superior representation of mobility patterns. This dataset includes detailed records such as Taxi ID, Timestamp, Latitude, Longitude, Occupancy Status, and Speed. For our analysis, we selected 36 taxis based on the continuity and consistency of their records, as well as the similarity in their active time periods. We use the 'Occupancy Status' to indicate task generation, where 1 (with passengers) represents a task and 0 (without passengers) indicates no task. The 'Speed' is used to categorize the task type based on the taxi's movement patterns. This experiment randomly selects one of the days in the Shenzhen. The simulation parameters are summarized in Table 4 in the Appendix.

**Baselines:** To emphasize the advantages of matching the degree of a task and a spatiotemporal unit in task delivery, we compare it with baselines. On the one hand, in order to verify the advantage of the STPWR method in predicting task generation based on publisher trajectory, we compared it with the following baselines: **Gibbs-Inspired LSTM**: [25] This method is developed by incorporating the principles used in Gibbs sampling to address the issues of high missing rate. **Unfilled LSTM**: This method does not adopt the filling method for prediction.

In order to verify the effectiveness of the workers' recruitment in STPWR method, we compared it with the following baselines: **Priority Greedy Method**: In this method, the urgency of a task is determined by its response time alone, with tasks having longer response times being given higher priority. **Distance Greedy Method**: In this method, the likelihood of a task being carried out is directly proportional to the brevity of the worker's movement distance, with shorter distances being more favorable. **Random Method**: In this method, recruitment is carried out in a random way without consideration of specific patterns.

Additionally, to verify the effectiveness of the worker pre-recruitment method, we will also compare the following methods: **OTAGA** [31]: In this method, worker recruitment is contingent upon the attributes of the task, without consideration for the pre-recruitment strategy. **GDRL** [29]: In this method, by integrating a Graph Attention Network (GAT) with Deep Reinforcement Learning (DRL) to address the worker recruitment problem without the premise of worker pre-recruitment.

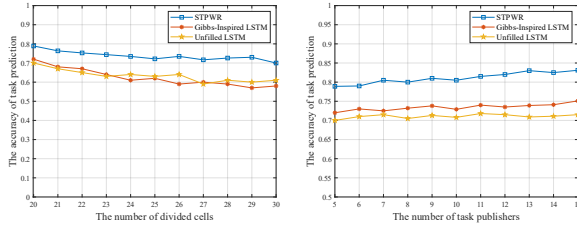


Figure 7: Comparison of task prediction accuracy across different numbers of cell divisions, task publishers.

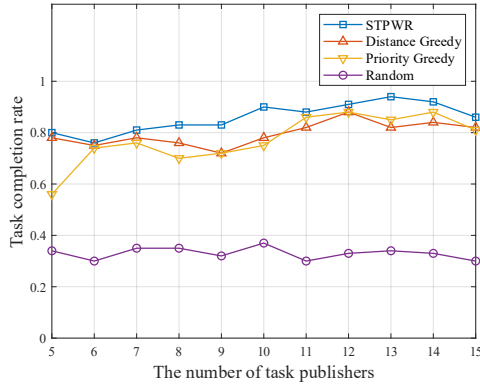


Figure 8: Comparison of task completion rates across methods by number of task publishers.

## 5.2 Results and Discussion

**The accuracy of prediction:** The accuracy of prediction is crucial when evaluating prediction methods. In this paper, we evaluate the accuracy of our prediction model by examining the difference between predicted and actual values. For this purpose, we use the accuracy of task generation prediction (i.e., the ratio of correctly predicted task generation events to the total actual task generation events) as our evaluation metric.

As shown in Figure 7, we analyze the impact of varying cells from  $20^2$  to  $30^2$  on the accuracy of task prediction. The results highlight the significant advantages of STPWR in accurately predicting task distribution. In comparison to the Gibbs-Inspired LSTM and the Unfilled LSTM, the advantage is attributed to our method's ability to leverage trajectories and behavior patterns of task publishers, leading to more reliable predictions. Figure 7 also represents the comparison of task prediction accuracy across different numbers of task publishers. The results reveal that the prediction accuracy of STPWR improves as the number of task publishers increases. This improvement is attributed to the method of filling missing information based on the similarity of trajectories and behavior patterns. As the number of task publishers increases, the availability of similar patterns enhances the accuracy of the predictions.

**Task Completion Rate (TCR):** It is a crucial criterion in worker pre-recruitment. We introduce a worker pre-recruitment method

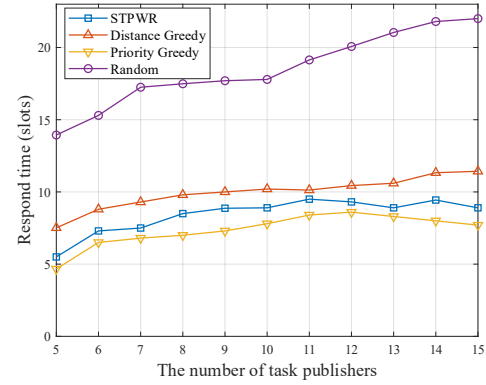


Figure 9: Comparison of task response time slots across methods by number of task publishers.

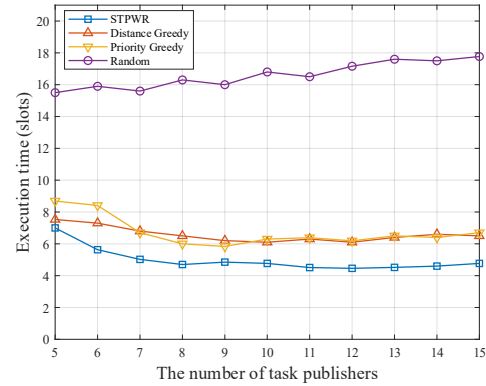


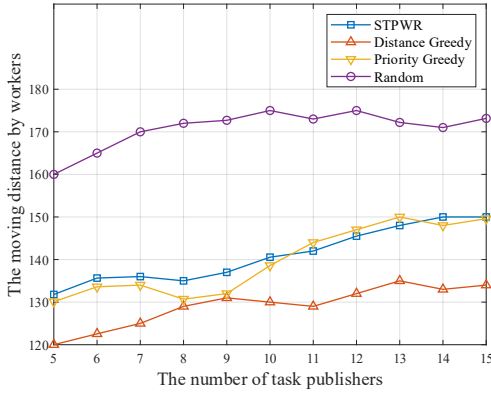
Figure 10: Comparison of task average execution time across methods by number of task publishers.

that considers various factors, including workers' activity scope and task priority.

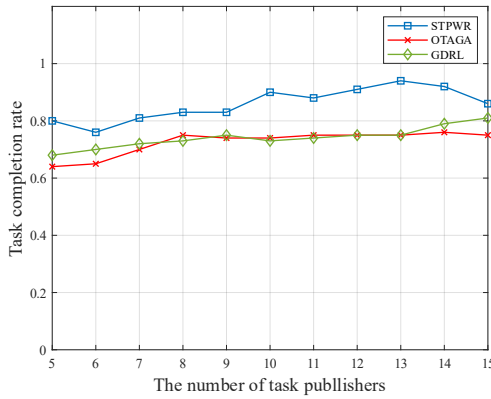
As shown in Figure 8, the task completion rate increases alongside the number of task publishers. The STPWR method demonstrates significant advantages, maintaining a high completion rate, which indicates high efficiency and stability in worker recruitment. In contrast, both the distance greedy and priority greedy methods show lower task completion rates compared to STPWR.

As shown in Figure 9, the priority greedy method exhibits the lowest task response time, while the distance greedy method performs the worst. In contrast, the STPWR method achieves a balanced performance by considering both worker movement distance and task response time.

In Figure 10, the comparison of the average execution time for completed tasks reveals that the STPWR method continues to demonstrate strong performance, maintaining a competitive edge



**Figure 11: Comparison of worker movement across methods by number of task publishers.**

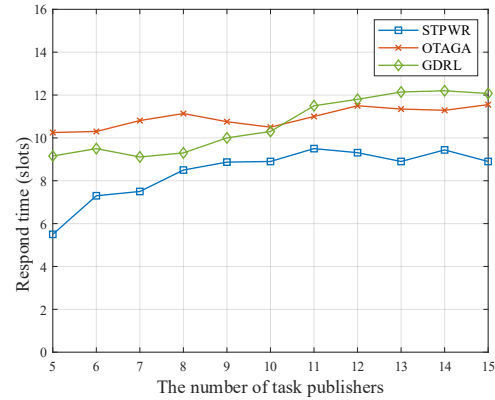


**Figure 12: Comparison of task completion rates across non-pre-recruitment methods by number of task publishers.**

despite executing a greater number of tasks. This suggests that the STPWR method exhibits robust efficiency and overall performance, even under higher operational demands.

In Figure 11, although the STPWR method is not optimal in optimizing the moving distance of workers, overall, STPWR still has certain advantages in optimizing workers' moving distance. This advantage is primarily due to the STPWR method's ability to execute more tasks without significantly increasing the workers' movement distance. This indicates that the STPWR algorithm can effectively balance task execution efficiency and worker movement distance in dynamic environments, ensuring that tasks are completed efficiently while reducing the burden of worker movement.

**The advantages of worker pre-recruitment:** In Figure 12, the STPWR method shows a clear advantage in task completion rate, while TOAGA and GDRL have lower rates. This difference is attributed to the STPWR method's consideration of factors such as workers' maximum mobility constraints and task types, enabling



**Figure 13: Comparison of task response time slots across non-pre-recruitment methods by number of task publishers.**

more precise worker recruitment. The method enhances task execution efficiency and improves overall task completion rates. By adjusting strategies, STPWR ensures optimal completion rates even in dynamic environments.

As illustrated in Figure 13, the STPWR method shows a clear advantage in reducing task responding time compared to approaches without worker pre-recruitment. This improvement arises from predicting task distribution, allowing timely worker recruitment. This proactive strategy minimizes response times and enhances task execution efficiency.

## 6 Conclusion And Discussion

This paper introduces the STPWR framework, aimed at improving the efficiency and sustainability of mobile crowd sensing systems. It enriches sparse historical data through a trajectory-filling technique that incorporates individual movement patterns and similarities among task publishers. The enriched data informs a predictive model using an LSTM architecture to forecast future movements and predict sensing tasks. Our worker pre-recruitment strategy utilizes multi-objective optimization to minimize task response delays and unnecessary worker movement, addressing operational efficiency and environmental concerns.

Future work includes adding fairness constraints to ensure equitable coverage in urban and rural areas, exploring advanced optimization techniques, and strengthening the theoretical foundations of pre-recruitment models. We also plan to validate our framework with humanitarian and urban management organizations in real-world scenarios like disaster response and environmental monitoring, aiming for more responsive, resource-efficient, and socially beneficial sensing systems.

## Acknowledgments

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## A Symbols And Descriptions

To formalize the interactions and computations within the proposed system model, we summarize the key mathematical notations in Table 1. These symbols are foundational to the problem formulation and the design of our STPWR method. The notations are categorized to reflect the core entities and relationships in our framework: spatiotemporal units, task publishers, workers, tasks, and the metrics for similarity and optimization. This structured overview aids in tracing the flow of information from the prediction of task distributions to the final recruitment decision variables. A clear grasp of these symbols is essential for comprehending the system's dynamics and replicating the experimental setup.

## B Description of Datasets

Table 2 shows the trajectory results of several task publishers after trajectory gridding. The table illustrates how raw location data is converted into a discrete sequence of grid cells over time. Empty entries in the table represent missing location records, highlighting the data sparsity issue inherent in real-world mobility datasets. This processed format serves as the primary input for the subsequent data completion and prediction stages of our framework.

**Table 1: Symbols And Descriptions**

| Symbol                       | Descriptions                                                                                |
|------------------------------|---------------------------------------------------------------------------------------------|
| $\mathbf{A}, A_h$            | Task publisher set and $h$ th task publisher                                                |
| $\mathbf{C}, C_m$            | Cell set and $m$ th cell                                                                    |
| $\mathbf{S}, S_i$            | Time slot set and $i$ th slot                                                               |
| $\mathbf{W}, W_k$            | Worker set and $k$ th worker                                                                |
| $\mathbf{G}, G_n$            | Task type set and $n$ th task type                                                          |
| $U_{i,m}$                    | The spatiotemporal unit within cell $C_m$ at time slot $S_i$                                |
| $\mathbf{T}, T_n^h(U_{i,m})$ | Task set and the task generated by task publisher $A_h$ in unit $U_{i,m}$ with type $G_n$   |
| $(x(U_{i,m}), y(U_{i,m}))$   | The coordinate of the center point of $U_{i,m}$                                             |
| $Cap(W_k)$                   | Execution capacity of worker $W_k$                                                          |
| $Pref(W_k)$                  | Task type preference of worker $W_k$                                                        |
| $Unit(W_k)$                  | The spatiotemporal unit $U_{i,m}$ of worker $W_k$                                           |
| $Cap(T_n^h(U_{i,m}))$        | The execution capacity required by task $T_n^h(U_{i,m})$                                    |
| $Cap(G_n)$                   | The execution capacity required by tasks with type $G_n$                                    |
| $WaitTime(T_n^h(U_{i,m}))$   | The response time of task $T_n^h(U_{i,m})$                                                  |
| $Cap'(T_n^h(U_{i,m}))$       | The total execution capacity of workers assigned to task $T_n^h(U_{i,m})$                   |
| $X_k(T_n^h(U_{i,m}))$        | The flag denoting whether task $T_n^h(U_{i,m})$ is assigned to worker $W_k$                 |
| $Dis(W_k)$                   | The movement distance of worker $W_k$                                                       |
| $P(A_h)(U_{i,m})$            | The probability of task publisher $A_h$ visiting $U_{i,m}$                                  |
| $P_n^h(G_n)(U_{i,m})$        | The probability of task publisher $A_h$ generating a task with type $G_n$ in unit $U_{i,m}$ |
| $Jaccard_{h,l}$              | The behavior similarity between task publishers $A_h$ and publisher $A_l$                   |
| $LCSS_{tr_h, tr_l}$          | The trajectory similarity between task publishers $A_h$ and publisher $A_l$                 |
| $I_h$                        | The historical task set of task publisher $A_h$                                             |

## C Chromosome Display

Table 3 provides an exposition of the chromosome binary encoding scheme. Each row represents a task, and each column corresponds to a worker. A binary value of ‘1’ indicates that the worker is pre-recruited for that specific task, while ‘0’ indicates otherwise. This encoding directly maps to a complete worker pre-recruitment plan and serves as the fundamental representation for all genetic operations within our optimization algorithm. By mapping these binary values to the decision variable  $X_k(T_n^h(U_{i,m}))$ , the framework can efficiently calculate the total execution capacity  $Cap'$  to evaluate the Task Completion Rate (TCR). Moreover, this matrix structure allows the fitness function to penalize excessive movement distances  $Dis(W_k)$  while prioritizing urgent tasks. As a result, genetic operations such as order crossover and mutation can effectively enhance the recruitment strategy, striking a balance between sensing quality and operational costs. To ensure practicality, the encoding includes constraints like the worker activity scope  $SCO(W_k)$ , preventing assignments to tasks outside their reachable range. Through iterative evolution, this representation enables the STPWR framework to converge towards an optimal recruitment configuration, reducing response delays in complex urban sensing environments.

**Table 2: Description of Datasets**

| DAY | TIME INTERVAL | AREA MOVEMENT |    |    |    |    |
|-----|---------------|---------------|----|----|----|----|
|     |               | P1            | P2 | P3 | P4 | P5 |
| d1  | 1             | c1            | c2 | c2 |    | c1 |
|     | 2             | c2            | c2 | c2 |    | c1 |
|     | 3             |               | c2 |    |    | c1 |
|     | 4             |               | c4 |    |    | c1 |
|     | 5             |               | c4 |    |    | c1 |
|     | 6             | c4            | c2 | c4 |    | c4 |
| d2  | 1             | c4            |    | c1 | c4 | c2 |
|     | 2             | c1            |    | c1 | c4 | c2 |
|     | 3             | c1            | c3 | c1 | c1 | c2 |
|     | 4             | c4            | c4 | c1 | c3 | c2 |
|     | 5             |               | c4 | c3 | c1 | c2 |
|     | 6             |               | c4 | c3 | c1 | c2 |
| d3  | 1             |               | c1 | c2 |    | c2 |
|     | 2             |               | c1 | c1 |    | c1 |
|     | 3             | c1            | c1 | c1 |    | c1 |
|     | 4             | c3            | c1 | c1 |    | c3 |
|     | 5             | c3            | c1 | c1 |    | c3 |
|     | 6             | c3            | c1 | c1 |    | c3 |
| d4  | 1             | c3            | c4 | c1 | c1 | c3 |
|     | 2             |               | c3 | c3 | c1 | c1 |
|     | 3             |               | c3 | c4 | c1 | c1 |
|     | 4             |               | c1 | c4 | c1 | c1 |
|     | 5             | c3            | c3 | c1 | c1 | c1 |
|     | 6             | c3            | c3 | c3 | c1 | c1 |

## D Algorithm Details

The algorithm delineates the comprehensive workflow of the Spatiotemporal Task Prediction-based Worker Pre-Recruitment (STPWR) framework. The algorithm takes as input the task publisher set  $A$ , the cell set  $C$ , the worker set  $W$ , and a periodization parameter  $\rho$ , and outputs the optimal worker pre-recruitment scheme. It begins by partitioning the sensing area into discrete grid cells. For each task publisher, their trajectory is processed through gridding and subsequently filled to mitigate data sparsity, leveraging both individual trajectory regularity and behavioral similarities among publishers. An LSTM-based model is then trained and employed to predict the spatiotemporal probability distribution of future task generation. Based on these predictions, a genetic algorithm is utilized to optimize the pre-recruitment strategy. This involves population initialization, followed by iterative selection, crossover, and mutation operations, ultimately converging to an optimal solution. The algorithm effectively balances multiple objectives, including task completion rate, response time, and worker travel distance, through this multi-objective optimization. The algorithm also incorporates real-time environmental factors and worker availability to enhance its predictive accuracy. By continuously updating the model with new data, it can adapt to dynamic conditions, ensuring that the pre-recruitment scheme remains effective over time.

**Algorithm 1** The spatiotemporal task prediction-based worker pre-recruitment (STPWR)

**Input:** The task publisher set  $A$ , cell set  $C$ , worker set  $W$ , Periodization  $\rho$

**Output:** The optimal worker pre-recruitment scheme

Divide the sensing area into  $M$  cells

**for**  $i = 1$  to  $|H|$  **do**

$A_h$  trajectory gridding

$A_h$  trajectory filling by trajectory regularity

$A_h$  trajectory filling by behavioral similarity

**end for**

Task prediction model training with  $\tau$

Set iteration counter  $t = 0$

**for**  $i = 1$  to  $|I|$  **do**

**for**  $j = 1$  to  $|H|$  **do**

        Input historical data to the task prediction model

        Predicting the  $P(A_h)(U_{i,m})$ ,

$P(P(T^h(U_{i,m})))|P(A_h)(U_{i,m}))$ ,

$P(P_n^h(G_n)(U_{i,m})|P(T^h(U_{i,m})))$

**Select the highest probability**

$P(P_n^h(G_n)(U_{i,m})|P(A_h)(U_{i,m}))$

**end for**

    Generate an initial candidate population  $POP_0$  with  $\mu$

**repeat**

$POP'_t \leftarrow$  select chromosome from the population using the selection operations  $POP_t$

$POP''_t \leftarrow$  generate new chromosome by order crossover between selected chromosomes on  $POP'_t$

$POP'''_t \leftarrow$  generate new chromosomes through mutation operations on  $POP''_t$

$POP_{t+1} \leftarrow POP'''_t$

$t = t + 1$

**until**  $t$  is the maximum number of iterations  $\gamma$

**if**  $\text{mod}(i, \rho) = 0$  **then**

        LSTM task prediction model training

**end if**

    Output the optimal worker pre-recruitment result for the current slot and the next slot

**end for**

**E Parameter Settings**

Table 4 summarizes the key hyperparameter settings utilized in our experiments. These parameters were determined via extensive grid search and cross-validation to ensure optimal model performance on the dataset. The parameters include: the number of task publishers  $H$ , the number of grid cells  $M$ , the number of time slots  $I$ , the number of task types  $N$ , the execution capacity requirement of a task  $Cap(T_n^h(U_{i,m}))$ , the weight parameter  $\eta$  for similarity calculation, the activity scope of workers  $SCO(W_k)$ , the execution capacity of a worker, the population size  $\mu$  for the genetic algorithm, the number of iterations  $\gamma$ , the sequence length  $\tau$  for the task prediction model, and the model training periodization  $\rho$ .

**Table 3: Display Chromosome**

|                  | $W_1$ | $W_2$ | $W_3$ | $W_4$ | $W_5$ | $W_6$ | $W_7$ | $W_8$ | $W_9$ | $W_{10}$ |
|------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|----------|
| $T_n^1(U_{i,m})$ | 1     | 1     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0        |
| $T_n^2(U_{i,m})$ | 0     | 0     | 1     | 1     | 0     | 0     | 0     | 1     | 1     | 1        |
| $T_n^3(U_{i,m})$ | 0     | 0     | 1     | 1     | 0     | 0     | 0     | 1     | 1     | 1        |
| $T_n^4(U_{i,m})$ | 0     | 0     | 0     | 0     | 1     | 0     | 1     | 0     | 0     | 0        |

**Table 4: Simulation Parameters**

| Symbol                | Description                                | Value   |
|-----------------------|--------------------------------------------|---------|
| $H$                   | The number of task publishers              | [5, 15] |
| $M$                   | The number of cells                        | 20×20   |
| $I$                   | The number of time slots                   | 288     |
| $N$                   | The number of task types                   | 5       |
| $Cap(T_n^h(U_{i,m}))$ | The execution capacity requirement of task | 15-20   |
| $\eta$                | The weight parameter                       | 05      |
| $SCO(W_k)$            | The scope of activity of workers           | 1-3     |
| $Cap(W_k)$            | Worker execution ability                   | 1-3     |
| $\mu$                 | Population size                            | 30      |
| $\gamma$              | Number of iterations                       | 30      |
| $\tau$                | Task prediction model sequence length      | 10      |
| $\rho$                | Model training periodization               | 20      |